

## Requirements:

### Question 1

1. Design the function `myFIRrecwin` without using built-in functions.
2. You have to write another `.m` file called `myFIRrecwin`, and then call the function in your main `.m` file
3. Plot the result of case 1.

### Question 2

1. Plot the result of case 2.
2. Compare the result to the one of case 1.
3. Describe and summarize the difference.

### Question 3

1. Design the Kaiser window by the specification of stop band tolerance  $\delta = 0.01$  and  $A = 40$ .
2. Derive  $\beta$  with  $A$  and  $\omega_c$  in the starter code.
3. Build the window function  $w$  with `besseli()`.
4. Plot the result of case 3.
5. Compare the result to the ones of case 2 and make detailed statements on the length difference with using difference windows.

### Question 4

1. Compare your previous result of HW5 (IIR Filtering) and your result of HW6 (FIR Filtering), determine which is more computationally efficient, and point out what factor makes the difference.
2. Discuss what other factors (change of parameters) will also affect the choice of using IIR or FIR.